

DAX Reference – Healthcare Claims Analytics

All DAX in this model, extracted directly from `ClaimsProject.pbix`.

Contents

- [Date Table](#)
- [Calculated Columns](#)
- [Measures](#)
- [Row-Level Security Roles](#)

Date Table

Calculated table with a July–June fiscal calendar. Marked as the model's date table; related to `Claims[ClaimDate]`.

```
Date = GENERATE (
    CALENDAR ( DATE ( 2022, 7, 1 ), DATE ( 2025, 6, 30 ) ),
    VAR d = [Date]
    VAR m = MONTH ( d )
    VAR fy = IF ( m >= 7, YEAR ( d ) + 1, YEAR ( d ) )
    VAR fm = MOD ( m - 7, 12 ) + 1
    VAR fq = ROUNDUP ( fm / 3, 0 )
    RETURN
        ROW (
            "Year", YEAR ( d ),
            "Month Number", m,
            "Month", FORMAT ( d, "MMM" ),
            "Month Year", FORMAT ( d, "MMM yyyy" ),
            "Month Year Sort", YEAR ( d ) * 100 + m,
            "Fiscal Year", "FY" & RIGHT ( fy, 2 ),
            "Fiscal Year Number", fy,
            "Fiscal Month Number", fm,
            "Fiscal Quarter", "Q" & fq,
            "Fiscal Year Quarter", "FY" & RIGHT ( fy, 2 ) & " Q" & fq,
            "Fiscal Year Quarter Sort", fy * 10 + fq,
            "Day of Week", FORMAT ( d, "ddd" ),
            "Day of Week Number", WEEKDAY ( d, 2 )
        )
    )
```

```
)  
)
```

Calculated Columns (Claims table)

Age Band

```
Age Band =  
SWITCH (  
    TRUE (),  
    Claims[PatientAge] < 18, "0-17",  
    Claims[PatientAge] < 35, "18-34",  
    Claims[PatientAge] < 50, "35-49",  
    Claims[PatientAge] < 65, "50-64",  
    "65+"  
)
```

Age Band Sort

```
Age Band Sort =  
SWITCH (  
    TRUE (),  
    Claims[PatientAge] < 18, 1,  
    Claims[PatientAge] < 35, 2,  
    Claims[PatientAge] < 50, 3,  
    Claims[PatientAge] < 65, 4,  
    5  
)
```

Income Band

```
Income Band =  
SWITCH (  
    TRUE (),  
    Claims[PatientIncome] < 40000, "Under $40K",  
    Claims[PatientIncome] < 70000, "$40K-$70K",  
    Claims[PatientIncome] < 100000, "$70K-$100K",  
    Claims[PatientIncome] < 130000, "$100K-$130K",
```

```
"$130K+"  
)
```

Income Band Sort

```
Income Band Sort =  
SWITCH (  
    TRUE (),  
    Claims[PatientIncome] < 40000, 1,  
    Claims[PatientIncome] < 70000, 2,  
    Claims[PatientIncome] < 100000, 3,  
    Claims[PatientIncome] < 130000, 4,  
    5  
)
```

Claim Size Tier

```
Claim Size Tier =  
SWITCH (  
    TRUE (),  
    Claims[ClaimAmount] < 2500, "Low (< $2.5K)",  
    Claims[ClaimAmount] < 7500, "Medium ($2.5K-$7.5K)",  
    "High ($7.5K+)"  
)
```

Is Denied

```
Is Denied =  
IF ( Claims[ClaimStatus] = "Denied", 1, 0 )
```

Days Pending

```
Days Pending =  
IF (  
    Claims[ClaimStatus] = "Pending",  
    DATEDIFF ( Claims[ClaimDate], DATE ( 2024, 7, 8 ), DAY ),  
    0  
)
```

Aging Bucket

```
Aging Bucket =  
IF (  
    Claims[ClaimStatus] = "Pending",  
    SWITCH (  
        TRUE (),  
        Claims[Days Pending] <= 30, "0-30 days",  
        Claims[Days Pending] <= 90, "31-90 days",  
        Claims[Days Pending] <= 180, "91-180 days",  
        Claims[Days Pending] <= 365, "181-365 days",  
        "Over 1 year"  
    )  
)
```

Aging Bucket Sort

```
Aging Bucket Sort =  
IF (  
    Claims[ClaimStatus] = "Pending",  
    SWITCH (  
        TRUE (),  
        Claims[Days Pending] <= 30, 1,  
        Claims[Days Pending] <= 90, 2,  
        Claims[Days Pending] <= 180, 3,  
        Claims[Days Pending] <= 365, 4,  
        5  
    )  
)
```

Data Quality Issue

```
Data Quality Issue =  
SWITCH (  
    TRUE (),  
    Claims[ProviderSpecialty] = "Pediatrics" && Claims[PatientAge] >= 18, "Adult seen in  
Pediatrics",  
    Claims[PatientEmploymentStatus] = "Employed" && Claims[PatientAge] < 14, "Employed under  
age 14",  
    Claims[PatientMaritalStatus] IN { "Married", "Widowed", "Divorced" } &&  
Claims[PatientAge] < 16, "Married/Widowed/Divorced under 16",  
    )
```

```
Claims[PatientEmploymentStatus] = "Retired" && Claims[PatientAge] < 50, "Retired under  
age 50",  
    "No issue"  
)
```

Measures (`_Measures` table)

Core Volume & Dollars

Total Claims

```
Total Claims =  
COUNTROWS ( Claims )
```

Total Billed

```
Total Billed =  
SUM ( Claims[ClaimAmount] )
```

Total Claims Amount

```
Total Claims Amount =  
Sum(Claims[ClaimAmount])
```

Avg Claim Amount

```
Avg Claim Amount =  
DIVIDE ( [Total Billed], [Total Claims] )
```

Claim Status

Approved Claims

```
Approved Claims =  
CALCULATE ( [Total Claims], Claims[ClaimStatus] = "Approved" )
```

Denied Claims

```
Denied Claims =  
CALCULATE ( [Total Claims], Claims[ClaimStatus] = "Denied" )
```

Pending Claims

```
Pending Claims =  
CALCULATE ( [Total Claims], Claims[ClaimStatus] = "Pending" )
```

Approved Amount

```
Approved Amount =  
CALCULATE ( [Total Billed], Claims[ClaimStatus] = "Approved" )
```

Denied Amount

```
Denied Amount =  
CALCULATE ( [Total Billed], Claims[ClaimStatus] = "Denied" )
```

Pending Amount

```
Pending Amount =  
CALCULATE ( [Total Billed], Claims[ClaimStatus] = "Pending" )
```

Rates & Revenue at Risk

Denial Rate

```
Denial Rate =  
DIVIDE ( [Denied Claims], [Total Claims] )
```

Adjudicated Denial Rate

```
Adjudicated Denial Rate =  
DIVIDE ( [Denied Claims], [Approved Claims] + [Denied Claims] )
```

Approval Rate

```
Approval Rate =  
DIVIDE ( [Approved Claims], [Total Claims] )
```

Pending Rate

```
Pending Rate =  
DIVIDE ( [Pending Claims], [Total Claims] )
```

Denied \$ Rate

```
Denied $ Rate =  
DIVIDE ( [Denied Amount], [Total Billed] )
```

Revenue at Risk

```
Revenue at Risk =  
[Denied Amount] + [Pending Amount]
```

Revenue at Risk %

```
Revenue at Risk % =  
DIVIDE ( [Revenue at Risk], [Total Billed] )
```

Time Intelligence (Fiscal Year ends June 30)

Billed FYTD

```
Billed FYTD =  
TOTALYTD ( [Total Billed], 'Date'[Date], "6/30" )
```

Billed PY

```
Billed PY =  
CALCULATE ( [Total Billed], SAMEPERIODLASTYEAR ( 'Date'[Date] ) )
```

Billed YoY %

```
Billed YoY % =  
VAR Curr = [Total Billed]  
VAR Prev = [Billed PY]  
RETURN  
    IF ( NOT ISBLANK ( Prev ), DIVIDE ( Curr - Prev, Prev ) )
```

Claims PY

```
Claims PY =  
CALCULATE ( [Total Claims], SAMEPERIODLASTYEAR ( 'Date'[Date] ) )
```

Claims YoY %

```
Claims YoY % =  
VAR Curr = [Total Claims]  
VAR Prev = [Claims PY]  
RETURN  
    IF ( NOT ISBLANK ( Prev ), DIVIDE ( Curr - Prev, Prev ) )
```

Denial Rate PY

```
Denial Rate PY =  
CALCULATE ( [Denial Rate], SAMEPERIODLASTYEAR ( 'Date'[Date] ) )
```

Denial Rate YoY Change (pts)

```
Denial Rate YoY Change (pts) =  
VAR Curr = [Denial Rate]  
VAR Prev = [Denial Rate PY]  
RETURN  
    IF ( NOT ISBLANK ( Prev ), ( Curr - Prev ) * 100 )
```

Billed Rolling 3M Avg

```
Billed Rolling 3M Avg =  
CALCULATE (   
    AVERAGEX ( VALUES ( 'Date'[Month Year Sort] ), [Total Billed] ),  
    DATESINPERIOD ( 'Date'[Date], MAX ( 'Date'[Date] ), -3, MONTH )  
)
```

Share & Ranking

% of Total Billed

```
% of Total Billed =  
DIVIDE ( [Total Billed], CALCULATE ( [Total Billed], ALLSELECTED ( ) ) )
```

% of Total Claims

```
% of Total Claims =  
DIVIDE ( [Total Claims], CALCULATE ( [Total Claims], ALLSELECTED ( ) ) )
```

Specialty Denial Rank

```
Specialty Denial Rank =  
IF (   
    HASONEVALUE ( Claims[ProviderSpecialty] ),  
    RANKX ( ALLSELECTED ( Claims[ProviderSpecialty] ), [Denial Rate], , DESC, DENSE )  
)
```

State Revenue at Risk Rank

```
State Revenue at Risk Rank =  
IF (   
    HASONEVALUE ( Claims[ProviderState] ),  
    RANKX ( ALLSELECTED ( Claims[ProviderState] ), [Revenue at Risk], , DESC, DENSE )  
)
```

Pending Claims Aging

Avg Days Pending

```
Avg Days Pending =  
CALCULATE (   
    AVERAGE ( Claims[Days Pending] ),  
    Claims[ClaimStatus] = "Pending"  
)
```

Pending Over 1 Year Amount

```
Pending Over 1 Year Amount =  
CALCULATE ( [Pending Amount], Claims[Aging Bucket] = "Over 1 year" )
```

Oldest Pending (Days)

```
Oldest Pending (Days) =  
MAX ( Claims[Days Pending] )
```

Data Quality

DQ Adult in Pediatrics

```
DQ Adult in Pediatrics =  
CALCULATE ( [Total Claims], Claims[ProviderSpecialty] = "Pediatrics", Claims[PatientAge] >=  
18 )
```

DQ Retired Under 50

```
DQ Retired Under 50 =  
CALCULATE ( [Total Claims], Claims[PatientEmploymentStatus] = "Retired", Claims[PatientAge] <  
50 )
```

DQ Minor with Adult Marital Status

```
DQ Minor with Adult Marital Status =  
CALCULATE (  
    [Total Claims],  
    Claims[PatientMaritalStatus] IN { "Married", "Widowed", "Divorced" },  
    Claims[PatientAge] < 16  
)
```

DQ Employed Under 14

```
DQ Employed Under 14 =  
CALCULATE ( [Total Claims], Claims[PatientEmploymentStatus] = "Employed", Claims[PatientAge]  
< 14 )
```

DQ Records with Any Issue

```
DQ Records with Any Issue =  
CALCULATE ( [Total Claims], Claims[Data Quality Issue] <> "No issue" )
```

DQ % of Records Affected

```
DQ % of Records Affected =  
DIVIDE ( [DQ Records with Any Issue], [Total Claims] )
```

DQ Amount Affected

```
DQ Amount Affected =  
CALCULATE ( [Total Billed], Claims[Data Quality Issue] <> "No issue" )
```

Clean Records

```
Clean Records =  
[Total Claims] - [DQ Records with Any Issue]
```

Dynamic Labels & Narrative

Top Diagnosis

```
Top Diagnosis =  
VAR TopDx =  
    TOPN ( 1, VALUES ( DimDiagnosis[DiagnosisDescription] ), [Total Claims], DESC )  
RETURN  
    CONCATENATEX ( TopDx, DimDiagnosis[DiagnosisDescription], ", " )
```

Top Procedure

```
Top Procedure =  
VAR TopPx =  
    TOPN ( 1, VALUES ( DimProcedure[ProcedureDescription] ), [Total Claims], DESC )  
RETURN  
    CONCATENATEX ( TopPx, DimProcedure[ProcedureDescription], ", " )
```

Selected Region Label

```
Selected Region Label =  
IF (   
    ISFILTERED ( Claims[ProviderRegion] ),  
    CONCATENATEX ( VALUES ( Claims[ProviderRegion] ), Claims[ProviderRegion], ", " ),  
    "All Regions"  
)
```

Selected FY Label

```
Selected FY Label =  
IF (   
    ISFILTERED ( 'Date'[Fiscal Year] ),
```

```

CONCATENATEX ( VALUES ( 'Date'[Fiscal Year] ), 'Date'[Fiscal Year], ", " ),
"All Years"
)

```

Title Denial Page

```

Title Denial Page =
"Denial Rate - " & [Selected Region Label] & " - " & [Selected FY Label]

```

Title Overview Page

```

Title Overview Page =
"Claims Overview - " & [Selected Region Label] & " - " & [Selected FY Label]

```

Denial Insight

```

Denial Insight =
VAR TopMethod =
    TOPN ( 1, VALUES ( Claims[ClaimSubmissionMethod] ), [Denial Rate], DESC )
VAR MethodName = MAXX ( TopMethod, Claims[ClaimSubmissionMethod] )
VAR MethodRate = MAXX ( TopMethod, [Denial Rate] )
RETURN
    FORMAT ( [Denied $ Rate], "0%" ) & " of billed dollars were denied. "
    & MethodName & " submissions have the highest denial rate ( "
    & FORMAT ( MethodRate, "0.0%" ) & " )."

```

Revenue at Risk Insight

```

Revenue at Risk Insight =
FORMAT ( [Revenue at Risk], "$#,0.0,,M" ) & " ( " & FORMAT ( [Revenue at Risk %], "0%" )
    & " ) of billed dollars are denied or still pending."

```

Row-Level Security Roles

All roles filter the `Claims` table.

Role	Filter expression
General Practice Director	[ProviderSpecialty] == "General Practice"
Neurology Director	[ProviderSpecialty] == "Neurology"
Orthopedics Director	[ProviderSpecialty] == "Orthopedics"
Pediatrics Director	[ProviderSpecialty] == "Pediatrics"
Dynamic Region Access	see below
Cardiology Director	[ProviderSpecialty] == "Cardiology"
Claims Director (All Access)	TRUE()

Dynamic Region Access

```

VAR UserRegions =
    CALCULATETABLE (
        VALUES ( UserSecurity[Region] ),
        UserSecurity[UserEmail] = USERPRINCIPALNAME ( )
    )
RETURN
    "All" IN UserRegions
    || [ProviderRegion] IN UserRegions

```